# EIG-MPHBS: a rotation-invariant information exchange for hybrid Honey Badger–Marine Predators optimisation

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**Abstract**

Hybrid metaheuristics that run two populations side by side often share information by recombining their members coordinate by coordinate. This exchange depends on the coordinate system: when the variables of a problem are coupled by a rotation, recombining along the original axes breaks the structure that each population has found. We study this weakness in MPHBS, a recent hybrid of the Honey Badger Algorithm and the Marine Predators Algorithm, and propose EIG-MPHBS, which performs the exchange in the eigenvector frame of the covariance of both populations in half of the iterations. The exchange step becomes provably invariant to rotations of the search space, uses exactly the same number of function evaluations as the original and adds 3.2–5.5% to the run time. We evaluate EIG-MPHBS in 18,840 runs on CEC2022 ($D = 10, 20$), the 24 BBOB functions ($D = 20$) and eight FIR filter-design problems, against MPHBS, its parent algorithms, four classical metaheuristics, L-SHADE and CMA-ES. In a test fixed before the experiments on 28 held-out instances, EIG-MPHBS improves on both MPHBS and its budget-matched random-mediator variant (Holm-adjusted $p = 0.043$); the margin is small and depends on how ties are treated. On all 56 instances it is never significantly worse than MPHBS, reduces the median error on the rotated ellipsoid 14.3-fold, and beats HBA, MPA, PSO, DE, GWO and WOA on 260 of 336 comparisons. It ranks third of eleven overall, behind L-SHADE and CMA-ES, and remains behind the CEC2022 entry EA4eig.

*Keywords:* Hybrid metaheuristics, Rotation invariance, Eigenvector crossover, Honey Badger Algorithm, Marine Predators Algorithm, FIR filter design

## 1. Introduction

Population-based metaheuristics are a standard tool for expert and intelligent systems that must optimise black-box objectives, from controller and filter design to scheduling and model tuning. A popular way to improve them is hybridisation: two algorithms with complementary search behaviour run together and exchange information (Talbi, 2002; Blum et al., 2011; Ma et al., 2019). MPHBS (Flores and Olivares, 2026) is a recent example. It runs a Honey Badger Algorithm (HBA) population (Hashim et al., 2022) and a Marine Predators Algorithm (MPA) population (Faramarzi et al., 2020) in parallel, and in every iteration a

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mediator assembles new candidate solutions dimension by dimension from the best members of both populations. MPHBS was proposed with a SARSA-style learner that chooses which population and which rank supplies each coordinate. A companion study (G, 2026) showed that a random choice at the same evaluation budget performs as well as the learner, and that the gains of MPHBS come from the exchange step itself.

This exchange has a structural weakness that it shares with many recombination operators: it acts on the coordinates of the search space. Copying coordinate $d$ from one solution into another is meaningful when the variables are separable, but when the objective couples variables through a rotation, coordinate-wise recombination destroys the correlated structure that the populations have learned (Salomon, 1996). In MPHBS this shows up clearly on rotated problems: on the BBOB rotated ellipsoid (f10), MPHBS ends with a median error of 121, while on its separable counterpart (f2) it reaches the $10^{-8}$ accuracy floor (Section 7.2).

Eigenvector-based recombination is a known remedy. Guo and Yang (2015) recombine differential evolution (DE) vectors in the eigenvector frame of the population covariance, CoBiDE (Wang et al., 2014) does the same with covariance matrix learning, EA4eig (Bujok and Kolenovský, 2022) applies eigen crossover inside a cooperative model of four algorithms, and in evolutionary multitasking Tang et al. (2021, 2022) transfer knowledge between populations in aligned eigen-subspaces. We do not propose a new operator. Our contribution is to identify the coordinate dependence of the HBA–MPA mediator as a cause of its weakness on rotated problems, to give the hybrid a rotation-invariant exchange that costs no extra evaluations, and to test the result rigorously against a broad set of algorithms. Specifically:

1. We propose EIG-MPHBS, which carries out the MPHBS exchange in the eigenvector frame of the pooled covariance of both populations in half of the iterations and along the original axes in the other half (Section 4).
2. We prove that the eigen-frame exchange is invariant to rotations of the search space and that it uses exactly the same number of function evaluations as the axis-frame exchange, at an extra arithmetic cost of $O(ND^2 + D^3)$ per iteration (Section 5).
3. We evaluate EIG-MPHBS on 56 instances from CEC2022, BBOB and an FIR filter-design application against ten algorithms: MPHBS and its random-mediator variant, the parent algorithms HBA and MPA, four widely used metaheuristics (PSO, DE, GWO, WOA), and two state-of-the-art references, L-SHADE and CMA-ES. We also compare with the official CEC2022 results of EA4eig (Section 7).
4. The main hypotheses, the held-out instances and the statistical tests were fixed in a design document before any confirmatory run. Code, raw results and the scripts that produce every number in this paper are public.

The results are positive but bounded, and we report both sides. EIG-MPHBS improves on MPHBS, is never significantly worse than it on any of the 56 instances, and clearly outperforms the common metaheuristics, including both of its parent algorithms. L-SHADE, CMA-ES and EA4eig remain stronger on most instances, and the confirmatory advantage on the held-out instances, while significant under the test we fixed in advance, is small.

The paper is organised as follows. Section 2 reviews related work, Section 3 describes MPHBS, Section 4 presents EIG-MPHBS and Section 5 its properties. Section 6 describes the experimental setup, Section 7 the results and Section 8 discusses them. Section 9 concludes.

## 2. Related work

### *2.1. HBA, MPA and their hybrids*

HBA (Hashim et al., 2022) alternates a digging move, scaled by a smell intensity, with a move towards the best agent, both modulated by a decreasing density factor. MPA (Faramarzi et al., 2020) switches between Brownian and Lévy movements in three phases and adds a marine memory and a fish-aggregating-device (FAD) perturbation. Both have many variants, hybrids and applications (Hassan et al., 2024; Al-Betar et al., 2023). HBA and MPA have been combined in series, population-division and switching schemes for power-network planning (Al-Dhaifallah et al., 2024); MPHBS couples them through a learned dimension-wise exchange (Flores and Olivares, 2026), which G (2026) showed to be no better than a budget-matched random choice.

### *2.2. Rotation invariance and coordinate-wise recombination*

Salomon (1996) showed that genetic algorithms with coordinate-wise operators lose much of their performance when a separable function is rotated, because their success relies on the independence of the variables. The same holds for binomial crossover in DE and for any operator that mixes solutions coordinate by coordinate. CMA-ES avoids the problem by adapting a full covariance matrix and is invariant to rotations of the search space (Hansen and Ostermeier, 2001); this is one reason why benchmark suites such as BBOB (Hansen et al., 2009, 2021) and CEC2022 (Kumar et al., 2021) include rotated functions.

### *2.3. Eigenvector-based recombination*

Several operators recombine solutions in a coordinate system that follows the population. Guo and Yang (2015) perform DE crossover in the eigenvector frame of the population covariance with a given probability, and CoBiDE uses covariance matrix learning in the same way (Wang et al., 2014). Civicioglu and Besdok (2018) propose a rotation-invariant crossover based on QR decomposition. EA4eig (Bujok and Kolenovský, 2022), a cooperative model of CMA-ES, CoBiDE, jSO and IDE with eigen crossover, was among the strongest entries of the CEC2022 competition. In evolutionary multitasking, Tang et al. (2021) project the individuals of two populations onto aligned eigen-subspaces before recombining them, and Tang et al. (2022) transfer knowledge between tasks in a coordinate system built from the major eigenvectors of both populations. QLRUN (Adam et al., 2027) combines Q-learning with covariance matrix learning in a single metaheuristic.

### *2.4. Positioning*

The idea of exchanging information between populations in an eigen frame of their joint distribution is therefore known; the closest precedents are the inter-task transfers of Tang et al. (2021, 2022). A Scopus search (title, abstract and keywords, 28 October 2026; queries in the repository) found no study that applies it to the dimension-wise exchange of HBA–MPA hybrids or that analyses the rotation dependence of such mediators. Our contribution is thus an application of a known principle to a new and recently popular class of hybrids, with a proof of its invariance in this setting, a demonstration of its cost, and a comparison against a wide set of algorithms under a design fixed in advance.

## 3. MPHBS

We minimise $f : \Omega \to \mathbb{R}$ over a box $\Omega = [\ell, u]^D$; points are row vectors. MPHBS maintains an HBA population $X^H \in \mathbb{R}^{N\times D}$ and an MPA population $X^M \in \mathbb{R}^{N\times D}$ ($N = 30$ each), their best members $p^H, p^M$, and the global best $p^\star$ with value $f^\star$. Each iteration has three stages (Algorithm 1).

*Phase 1: native moves.* HBA and MPA each update their population with their own operators; each candidate is evaluated and replaces its parent if better, and MPA applies its marine-memory rule. A FAD perturbation of the MPA population follows, before Phase 2 on CEC2022 and BBOB and after it on FIR, as in the authors' configurations. Phase 1 costs $2N$ evaluations, the FAD stage $N$. A full description of the operators is given by Flores and Olivares (2026) and G (2026).

*Phase 2: the mediator.* From each population the mediator takes a subset of $N_{\text{sub}} = \min(N, \max(5, \text{round}(\rho N)))$ members (half of them the best, the rest at random), sorted by fitness, and stacks them into a tensor $R \in \mathbb{R}^{N_{\text{sub}}\times D\times 2}$, where $R_{n,d,c}$ is coordinate $d$ of the rank-$n$ member of source $c \in \{H, M\}$. Starting from $v = p^\star$ (one evaluation), it runs $E = N_{\text{sub}} N_i$ episodes. The dimensions are split into five blocks of size $\lceil D/5 \rceil$ visited along random permutations. In episode $e$ with active block $A_e$, $K$ candidates are built: candidate $k$ copies $v$ and, for each $d \in A_e$, sets $v'_d = R_{n_d,d,c_d}$ for a chosen source $c_d$ and rank $n_d$. In MPHBS the choice is made by policies over a SARSA value table; in the random-mediator variant MPHBS-R it is drawn uniformly. A proxy score that needs no evaluation,

$$s(v') = -w_{\text{rank}}\,\overline{(n_d + 1)}_{d\in A_e} + w_{\text{nov}}\,\overline{|v'_d - v_d|}_{d\in A_e} - w_{\text{best}}\,\|v'_{A_e} - p^\star_{A_e}\|_2, \tag{1}$$

selects one candidate, which is evaluated (one evaluation per episode). It is accepted if it improves $f(v)$, and otherwise with a small probability that decreases over the run. Accepted candidates replace the worst subset members that they improve, and a new global best is copied into both subsets. At the end of Phase 2 the subsets are written back.

*Configurations.* We use the authors' configurations: $K = 7$, $N_i = 3$, $\rho = 0.25$ (so $N_{\text{sub}} = 8$, $E = 24$) on CEC2022 and BBOB, and $K = 3$, $N_i = 9$ ($E = 72$) on FIR, with $w_{\text{rank}} = 1$, $w_{\text{nov}} = 0.3$, $w_{\text{best}} = 0.05$. One iteration costs $3N + 1 + E$ evaluations.

## 4. EIG-MPHBS

### *4.1. Why the exchange depends on the coordinate system*

The mediator builds each candidate by taking coordinate $d$ from one stored solution and coordinate $d'$ from another. If the objective is separable, this is a sound way to combine good partial solutions. If the objective is $f(x) = g(xQ)$ for a rotation $Q$, the meaningful directions are the columns of $Q$, and a candidate assembled from the coordinates of different solutions generally lies far from all of them along those directions. Its value is then poor even when every source solution is good, so few exchange candidates are accepted and the exchange contributes little. The populations themselves carry the information needed to correct this: on such problems their spread aligns with the shape of the basin they occupy, so the principal axes of their covariance approximate the directions of the problem.

**Algorithm 1** One iteration of EIG-MPHBS. Lines 4–9 are the only differences from MPHBS-R.

**Input:** populations $X^H, X^M$, global best $p^\star$, iteration $t$, eigen-frame probability $q$

1: HBA and MPA moves; evaluate; greedy replacement; marine memory ▷ $2N$ FEs
2: **if** FAD before Phase 2 **then** FAD stage ▷ $N$
3: $p^\star \leftarrow$ best of $p^H, p^M$
4: **if** $U(0,1) < q$ **then** $B \leftarrow$ eigenvectors of $\mathrm{cov}([X^H; X^M])$ **else** $B \leftarrow I$ ▷ 0 FEs
5: draw ranked subsets $S^H, S^M$; $R \leftarrow$ projected subsets $S^H B$, $S^M B$; $z \leftarrow p^\star B$; $f_z \leftarrow f(p^\star)$ ▷ 1
6: **for** $e = 1, \ldots, E$ **do**
7: $\quad A_e \leftarrow$ next block of dimensions
8: $\quad$ for $k = 1..K$: $z^{(k)} \leftarrow z$; for $d \in A_e$ draw $(c_d, n_d)$ uniformly; $z_d^{(k)} \leftarrow R_{n_d,d,c_d}$
9: $\quad z' \leftarrow \arg\max_k s(z^{(k)})$ (1); $x' \leftarrow \mathrm{clip}(z'B^\top)$; $f' \leftarrow f(x')$ ▷ 1
10: $\quad$ **if** accepted **then** $z \leftarrow x'B$, $f_z \leftarrow f'$; update $S^H, S^M$ and their projections in $R$
11: $\quad$ **if** $f' < f^\star$ **then** $p^\star \leftarrow x'$; copy $x'$ into a random row of $S^H$ and $S^M$
12: write $S^H, S^M$ back into $X^H, X^M$; update $p^H, p^M, p^\star$
13: **if** FAD after Phase 2 **then** FAD stage ▷ $N$

*4.2. The eigen-frame exchange*

At the start of Phase 2, with probability $q$, EIG-MPHBS computes the sample covariance of the pooled populations,

$$C = \mathrm{cov}\begin{pmatrix} X^H \\ X^M \end{pmatrix} = B\Lambda B^\top, \tag{2}$$

with an orthonormal eigenvector matrix $B$ (columns ordered by eigenvalue). All mediator operations of the iteration are then performed on the coordinates $z = xB$ instead of $x$: the tensor $R$ holds the projected subsets, the start point is $z = p^\star B$, candidates are assembled coordinate-wise in $z$, and the proxy score (1) is computed in $z$. The selected candidate is mapped back and clipped to the box, $x' = \mathrm{clip}(z'B^\top)$, and evaluated; if it is accepted, its eigen coordinates are recomputed from the clipped point, $z \leftarrow x'B$. With probability $1 - q$ the iteration uses the original axes ($B = I$), so the method reduces to MPHBS-R in those iterations.

EIG-MPHBS uses the random mediator of MPHBS-R, not the SARSA learner, because the learner brings no benefit at the same budget (G, 2026) and its value table is indexed by coordinates, whose meaning would change whenever the frame changes. We set $q = 0.5$. This keeps the axis frame available for separable or nearly separable structure, on which coordinate-wise exchange is effective, while using the eigen frame for coupled structure. The value $q = 0.5$ was chosen over $q = 1$ in a 10-run pilot on 32 instances (BBOB f1, f3, f8, f10, f12, f15, f16, f21; CEC2022 $D = 20$; FIR); those instances are excluded from the confirmatory test (Section 6.3). Algorithm 1 summarises one iteration and Fig. 1 the exchange.

## 5. Properties

**Proposition 1** (Rotation invariance of the exchange)**.** *Let $Q$ be an orthogonal matrix and consider the rotated problem $\tilde{f}(y) = f(yQ^\top)$. Suppose the populations, subsets and global best of the rotated run are the rotated images $\tilde{X} = XQ$ of those of the original run, the pooled*

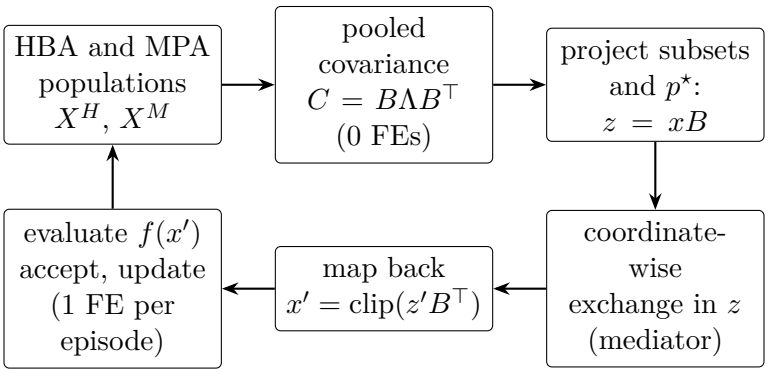



Figure 1: The eigen-frame exchange of EIG-MPHBS in one iteration (used with probability $q = 0.5$; otherwise $B = I$).

*covariance has distinct eigenvalues, the same random numbers are drawn, and no clipping occurs. Then the eigen-frame exchange produces, in every episode, the candidate $\tilde{x}' = x'Q$, with the same objective value and the same acceptance decision.*

*Proof.* The pooled covariance of the rotated populations is $\tilde{C} = Q^\top CQ$, which has the same eigenvalues as $C$ and, since they are distinct, eigenvectors $\tilde{B} = Q^\top BS$ with $S$ a diagonal matrix of signs, in the same order. The eigen coordinates of a rotated point are therefore $\tilde{z} = xQ\,Q^\top BS = zS$: each coordinate equals the original one up to a fixed sign. Ranks depend only on objective values, which are equal because $\tilde{f}(xQ) = f(x)$. Assembling a candidate by copying coordinate $d$ from a stored solution commutes with multiplication by $S$, so $\tilde{z}' = z'S$. The terms of the proxy score (1) are a mean rank, a mean of absolute coordinate differences and a Euclidean norm, all unchanged by sign flips, so the same candidate is selected. Mapping back gives $\tilde{x}' = \tilde{z}'\tilde{B}^\top = z'SS^\top B^\top Q = x'Q$, whose value is $\tilde{f}(x'Q) = f(x')$, and the acceptance rule depends only on objective values. □

*Remark* 1. The axis-frame exchange ($B = I$) has no such property: rotating the problem changes which coordinates are copied together. Proposition 1 concerns the exchange step only. The HBA and MPA moves of Phase 1 are coordinate-wise and not rotation invariant, and neither is clipping to the box, so EIG-MPHBS as a whole is not rotation invariant. A unit test in the repository checks the invariance of the eigen coordinates numerically.

**Proposition 2** (No extra evaluations)**.** *In every iteration EIG-MPHBS performs exactly the same number of function evaluations as MPHBS-R, $3N + 1 + E$. The additional arithmetic cost is $O(ND^2 + D^3)$ for the covariance and its eigendecomposition and $O(D^2)$ per episode for the projections, i.e. $O(ND^2 + D^3 + ED^2)$ per iteration.*

*Proof.* The frame is computed from points already evaluated, and the mediator evaluates one start point and one candidate per episode in either frame. The covariance of $2N$ points costs $O(ND^2)$, the symmetric eigendecomposition $O(D^3)$, and each projection or back-projection of a $D$-vector $O(D^2)$. □

For $D \leq 31$ and $N = 30$ this overhead is negligible compared with the cost of the objective evaluations and the mediator itself; Section 7.5 measures it. A unit test checks that EIG-MPHBS and MPHBS-R use identical evaluation counts on every run.

## 6. Experimental setup

### *6.1. Benchmarks*

We use four benchmark sets (56 instances):

- CEC2022 (Kumar et al., 2021), F1–F12, at $D = 10$ and $D = 20$ (24 instances), computed with the official code.
- BBOB (Hansen et al., 2009), f1–f24 at $D = 20$, with problem instance $((r-1) \bmod 15)+1$ in run $r$. The suite contains separable and rotated versions of the same functions, which allows a direct test of rotation dependence.
- Eight FIR filter-design problems from the MPHBS study (Flores and Olivares, 2026) ($D = 31$): low-pass, high-pass, band-pass and band-stop filters, each in an easier and a harder variant, with coefficients $h \in [-1, 1]^{31}$ and the objective

$$F(h) = W_p \frac{1}{|P|} \sum_{n \in P} \Big( |H(\omega_n)| - 1 \Big)^2 + W_s \frac{1}{|S|} \sum_{n \in S} |H(\omega_n)|^2, \tag{3}$$

  where $H$ is the 2048-point discrete Fourier transform of $h$, $P$ and $S$ are the pass- and stop-band grid points, $W_p = 1$ and $W_s \in \{100, 120\}$.

The budget is $2 \times 10^5$ evaluations on CEC2022 and BBOB and $1.5 \times 10^5$ on FIR. Each algorithm is run 30 times per instance. The outcome is the final error $f(x_{\text{best}}) - f^\star$ (the objective value on FIR), floored at $10^{-8}$.

### *6.2. Algorithms*

Table 1 lists the eleven algorithms. MPHBS, MPHBS-R and EIG-MPHBS share all code except the flags shown. HBA and MPA are the parent algorithms as ported in our framework, which was validated against the MPHBS authors' MATLAB results (no significant differences in 80 distribution comparisons on CEC2022 and FIR). PSO, DE, GWO and WOA are four widely used metaheuristics with their published default settings, and L-SHADE (Tanabe and Fukunaga, 2014) and CMA-ES with IPOP restarts (Hansen and Ostermeier, 2001; Auger and Hansen, 2005; Hansen et al., 2019) are state-of-the-art references. All algorithms use the same budget; the classical ones stop exactly at the budget, while the MPHBS family stops after the last complete iteration (199 930 of 200 000 evaluations on CEC2022 and BBOB, 149 857 of 150 000 on FIR). No algorithm was tuned on our benchmarks beyond the choice of $q$ described above.

### *6.3. Design and statistics*

Runs with the same benchmark, instance and run number share their seed across all algorithms, so comparisons are paired. Before any confirmatory run we fixed in a design document the arms, the benchmarks and two hypotheses on a held-out set of 28 instances that were not used in any pilot: CEC2022 $D = 10$ F1–F12 and BBOB f2, f4–f7, f9, f11, f13, f14, f17–f20, f22–f24. For each instance we compute the Vargha–Delaney effect size (Vargha and Delaney, 2000) $A = P(e_1 < e_2) + \frac{1}{2}P(e_1 = e_2)$ over all $30 \times 30$ pairs of final errors, where $e_1$ belongs to EIG-MPHBS.

Table 1: Algorithms and settings. $N$: population size; HBA–MPA hybrids run two populations of $N$ each.

| Algorithm | Settings |
|---|---|
| EIG-MPHBS | proposed; MPHBS with random mediator and eigen-frame exchange with probability $q = 0.5$; $N = 30$ |
| MPHBS-R | MPHBS with a uniform random mediator at the same budget (G, 2026); $N = 30$ |
| MPHBS | as published, SARSA mediator (Flores and Olivares, 2026); $N = 30$; $K, N_i, \rho$ as in Section 3 |
| HBA | Hashim et al. (2022); $N = 30$, $\beta = 6$, $C = 2$ |
| MPA | Faramarzi et al. (2020); $N = 30$, $P = 0.5$, FADs $= 0.2$ |
| PSO | global best, inertia $0.9 \to 0.4$ (Kennedy and Eberhart, 1995; Shi and Eberhart, 1998); $c_1 = c_2 = 2$, $\lvert v\rvert \le 0.2(u - \ell)$; $N = 30$ |
| DE | DE/rand/1/bin (Storn and Price, 1997); $F = 0.5$, $CR = 0.9$; $N = 30$ |
| GWO | Mirjalili et al. (2014); $a : 2 \to 0$; $N = 30$ |
| WOA | Mirjalili and Lewis (2016); $a : 2 \to 0$, $b = 1$; $N = 30$ |
| L-SHADE | Tanabe and Fukunaga (2014); $N_{\text{init}} = 18D$, $N_{\min} = 4$, $H = 6$, archive rate 2.6, $p = 0.11$ |
| CMA-ES | IPOP-CMA-ES (Auger and Hansen, 2005), pycma (Hansen et al., 2019); $\sigma_0 = 0.3(u - \ell)$, default $\lambda$, population doubled at each restart |

- **H1**: EIG-MPHBS outperforms MPHBS-R, i.e. the median of $A$ over the held-out instances exceeds 0.5. This isolates the effect of the eigen frame.
- **H2**: EIG-MPHBS outperforms MPHBS.

Each is tested with a one-sided Wilcoxon signed-rank test (Wilcoxon, 1945) of $A - 0.5$ over the 28 instances, with Holm correction (Holm, 1979) over the two hypotheses at $\alpha = 0.05$. All other analyses are secondary and descriptive: the same statistics on all 56 instances; per-instance paired Wilcoxon tests with Holm correction within each benchmark; Friedman ranks (Derrac et al., 2011; Demšar, 2006); the rotation pairs of BBOB; the comparison with EA4eig; and run time. The six classical algorithms were added after the main analysis to provide the comparisons that are customary in this field; their addition is recorded in the design document and does not affect H1 and H2.

## 7. Results

### *7.1. Confirmatory tests*

Table 2 and Fig. 2 show the confirmatory results. On the held-out instances, EIG-MPHBS is better than MPHBS-R on 14 instances, tied on 6 and worse on 8 (one-sided $p = 0.031$), and better than MPHBS on 15, tied on 5 and worse on 8 ($p = 0.022$). After Holm correction both hypotheses are supported ($p_{\text{Holm}} = 0.043$ for each). The evidence is not strong, however. The median effect is small (0.500 and 0.514), because most CEC2022 $D = 10$ functions are solved equally well by all three variants, and when the instances with $A = 0.5$ are kept in the test with Pratt or zero-split handling the Holm-adjusted $p$-values become 0.056 and 0.057.

Table 2: Confirmatory tests on the 28 held-out instances (H1, H2) and their repetition on all 56 instances (H1′, H2′, secondary: the 28 additional instances include those used to choose the eigen-frame probability). $\tilde{A}$: median Vargha–Delaney effect over instances ($> 0.5$ favours EIG-MPHBS). W/T/L: instances with $A > 0.5$, $A = 0.5$, $A < 0.5$. $p$: one-sided Wilcoxon signed-rank test of $A - 0.5$ over instances; Holm over {H1, H2}. The two sensitivity columns keep the instances with $A = 0.5$ (Pratt and zero-split handling). Sig.: instances where a paired per-instance test (Holm within benchmark) favours EIG-MPHBS / the comparator.

| | Comparison | $\tilde{A}$ | W/T/L | $p$ | $p_{\text{Holm}}$ | $p_{\text{Holm}}$ (Pratt) | $p_{\text{Holm}}$ (zsplit) | Sig. |
|---|---|---|---|---|---|---|---|---|
| H1 | EIG-MPHBS vs MPHBS-R | 0.500 | 14/6/8 | 0.031 | 0.043 | 0.056 | 0.057 | 5/1 |
| H2 | EIG-MPHBS vs MPHBS | 0.514 | 15/5/8 | 0.022 | 0.043 | 0.056 | 0.057 | 6/0 |
| H1′ | all 56 instances (vs MPHBS-R) | 0.520 | 31/7/18 | $7.7\times10^{-4}$ | – | – | – | 12/1 |
| H2′ | all 56 instances (vs MPHBS) | 0.529 | 33/6/17 | $5.2\times10^{-4}$ | – | – | – | 14/0 |

The decision rule fixed in advance used the default handling, which drops these instances; we report the alternatives so that readers can weigh the result.

The per-instance tests show where the effect comes from. Against MPHBS-R, EIG-MPHBS is significantly better on 5 held-out instances (BBOB f6, f9, f11, f14, f19) and worse on 1 (f4, the separable Rastrigin with asymmetric penalty). Against MPHBS it is better on 6 and never significantly worse. The gains are concentrated on BBOB functions with ill-conditioned or rotated structure.

On all 56 instances, including the 28 used in the pilot or not held out, the picture is clearer: the median $A$ is 0.520 against MPHBS-R ($p = 7.7\times10^{-4}$; 12 instances significantly better, 1 worse) and 0.529 against MPHBS ($p = 5.2\times10^{-4}$; 14 better, 0 worse). The largest effects are on FIR (median $A$ 0.703 against MPHBS) and CEC2022 $D = 20$ (0.611). These instances include the pilot instances on which $q$ was chosen, so their effect sizes may be optimistic.

### *7.2. Rotation dependence*

Table 3 compares BBOB functions that differ only by a rotation. On the separable ellipsoid (f2) all algorithms reach the accuracy floor. On the rotated ellipsoid (f10), MPHBS and MPHBS-R stall at median errors of 121 and 89.9, while EIG-MPHBS reaches 8.47, a 10.6-fold reduction relative to MPHBS-R. This is the mechanism the method targets: the exchange in the eigen frame recombines along the principal axes of the ellipsoid. The gap to CMA-ES and L-SHADE, which solve f10, remains, because Phase 1 of the hybrid is still coordinate-wise. On Rastrigin, the rotated version (f15) is hard for all three hybrids and the eigen frame does not help (median 54.2 against 58.7 for MPHBS-R); on the separable version (f3) EIG-MPHBS is slightly worse (1.99 against 0.995), as expected when half of the exchanges no longer use the axes along which the function separates. Multimodal rotated problems need more than a change of frame in the exchange step.

### *7.3. Comparison with ten algorithms*

Table 4 gives the mean Friedman ranks of the eleven algorithms and Table 5 the per-instance significance counts of EIG-MPHBS against each of them; Tables A.8–A.11 in Appendix A give the median errors per instance. The Friedman test rejects equal performance on every benchmark ($p \leq 1.4\times10^{-12}$).

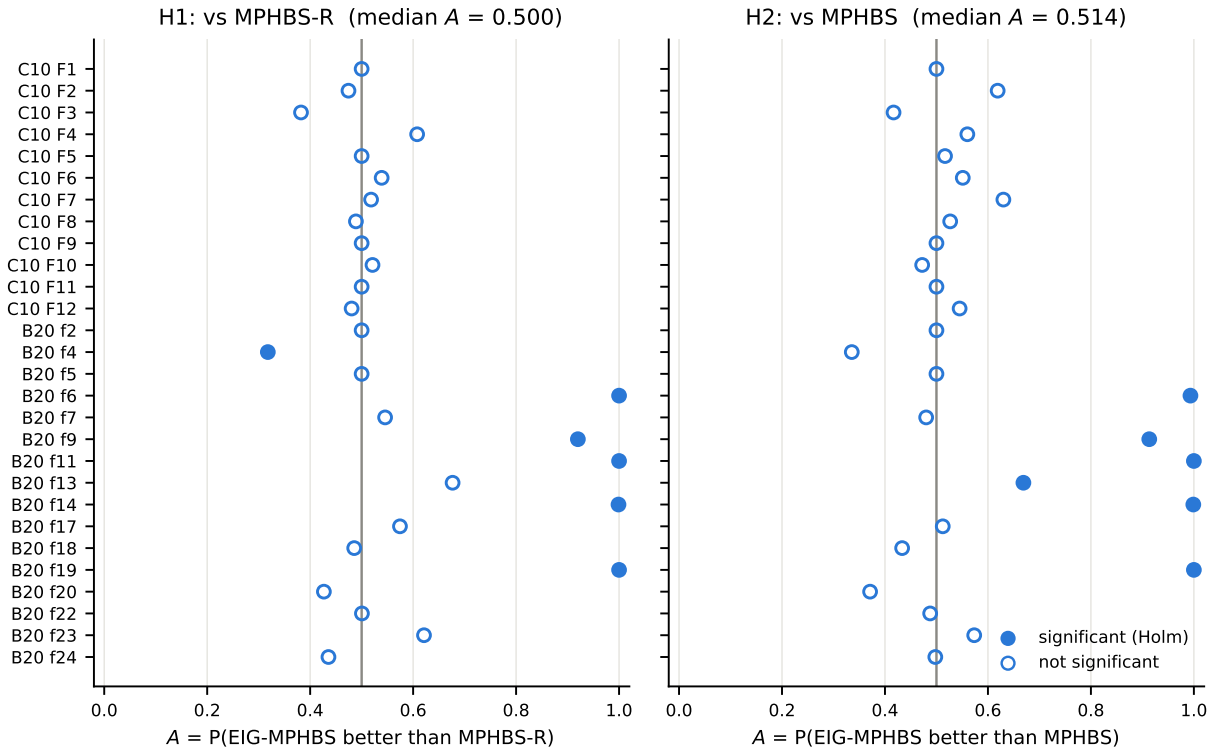



Figure 2: Effect size $A$ of EIG-MPHBS on each of the 28 held-out instances against MPHBS-R (H1) and MPHBS (H2); $A > 0.5$ favours EIG-MPHBS. Filled markers: significant per-instance paired test (Holm within benchmark). C10: CEC2022 $D = 10$; B20: BBOB $D = 20$.

EIG-MPHBS ranks 3 of 11 overall (mean rank 4.01), behind L-SHADE (2.60) and CMA-ES (2.97) and ahead of MPHBS-R (4.62), MPHBS (4.64) and all classical metaheuristics. It has the best mean rank on CEC2022 $D = 10$ and ranks third on the other three benchmarks.

- **Against its own family**, EIG-MPHBS is significantly better than MPHBS on 14 instances and worse on none, and better than MPHBS-R on 12 and worse on 1.
- **Against the parent algorithms**, it is significantly better than HBA on 47 of 56 instances and worse on none, and better than MPA on 24 and worse on 6; MPA is the strongest of the classical algorithms, and many of the ties with it are CEC2022 $D = 10$ functions that both solve.
- **Against PSO, DE, GWO and WOA**, it is better on 44, 37, 52 and 56 instances and worse on 1, 6, 0 and 0. Over the six classical algorithms it wins 260 and loses 13 of 336 comparisons.
- **Against L-SHADE and CMA-ES**, it is better on 4 and 8 instances and worse on 34 and 31. On CEC2022 $D = 10$ the three are close, but at $D = 20$, on BBOB and on FIR the two references are clearly stronger.

GWO and WOA perform poorly on these shifted problems; both are known to be biased

Table 3: Median final error on BBOB function pairs that differ only by a rotation ($D = 20$, 30 runs; values at or below $10^{-8}$ shown as 0).

| | Ellipsoid | | Rastrigin | |
|---|---|---|---|---|
| Algorithm | f2 (separable) | f10 (rotated) | f3 (separable) | f15 (rotated) |
| EIG-MPHBS | 0 | 8.47 | 1.99 | 54.2 |
| MPHBS-R | 0 | 89.9 | 0.995 | 58.7 |
| MPHBS | 0 | 121 | 0.995 | 51.2 |
| CMA-ES | 0 | 0 | 8.95 | 2.98 |
| L-SHADE | 0 | 0 | 0 | 4.13 |

Table 4: Mean Friedman rank of the eleven algorithms (rank of the median final error on each instance, 1 = best; lowest in bold), per benchmark and over all 56 instances. Last row: Friedman test $p$ per benchmark.

| Algorithm | C10 (12) | C20 (12) | B20 (24) | FIR (8) | All (56) |
|---|---|---|---|---|---|
| L-SHADE | 3.67 | **1.71** | **2.71** | 2.00 | **2.60** |
| CMA-ES | 4.92 | 2.38 | 2.96 | **1.00** | 2.97 |
| EIG-MPHBS | **3.54** | 4.17 | 4.38 | 3.38 | 4.01 |
| MPHBS-R | 3.79 | 4.92 | 5.00 | 4.25 | 4.62 |
| MPHBS | 4.50 | 4.83 | 4.71 | 4.38 | 4.64 |
| MPA | 3.58 | 5.67 | 4.73 | 6.12 | 4.88 |
| DE | 5.83 | 6.33 | 6.19 | 7.88 | 6.38 |
| HBA | 8.12 | 7.92 | 7.54 | 7.00 | 7.67 |
| PSO | 7.71 | 7.75 | 8.17 | 9.88 | 8.22 |
| GWO | 9.50 | 9.33 | 9.04 | 9.50 | 9.27 |
| WOA | 10.83 | 11.00 | 10.58 | 10.62 | 10.73 |
| Friedman $p$ | $3.9\times10^{-14}$ | $1.3\times10^{-14}$ | $2.8\times10^{-26}$ | $1.4\times10^{-12}$ | – |

towards the centre of the search space (Kůdela, 2022), which our unit tests confirm on a shifted sphere.

Fig. 3 shows median convergence curves of seven algorithms on six instances. On BBOB f10, f14 and f19 and on CEC2022 $D = 20$ F1, EIG-MPHBS separates from MPHBS and MPHBS-R early and keeps its lead. On f4 the eigen frame slows it down slightly, and on the FIR case shown all three hybrids converge to similar values, with EIG-MPHBS ahead in mid-run. CMA-ES converges fastest on most instances.

### *7.4. Comparison with EA4eig*

EA4eig (Bujok and Kolenovský, 2022) combines eigen crossover with four strong algorithms and was among the best entries of the CEC2022 competition. We compare EIG-MPHBS with the official EA4eig results at the competition budgets (30 runs at $2 \times 10^5$ evaluations for $D = 10$ and $10^6$ for $D = 20$; EIG-MPHBS was rerun at $10^6$ for $D = 20$). The comparison is unpaired and descriptive, because hardware, language and implementation differ. Table 6 shows that EA4eig is significantly better on 8 of 12 functions at $D = 10$ and 10 of 12 at $D = 20$, with no significant difference on the rest; EIG-MPHBS is never significantly

Table 5: EIG-MPHBS against each algorithm: number of instances where it is significantly better / not significantly different / significantly worse (paired two-sided Wilcoxon signed-rank test over the 30 seed-paired runs, Holm within each benchmark and comparator, $\alpha = 0.05$).

| vs | C10 (12) | C20 (12) | B20 (24) | FIR (8) | Total (56) |
|---|---|---|---|---|---|
| L-SHADE | 1/9/2 | 0/4/8 | 3/5/16 | 0/0/8 | 4/18/34 |
| CMA-ES | 3/7/2 | 1/5/6 | 4/5/15 | 0/0/8 | 8/17/31 |
| MPHBS-R | 0/12/0 | 3/9/0 | 7/16/1 | 2/6/0 | 12/43/1 |
| MPHBS | 0/12/0 | 2/10/0 | 7/17/0 | 5/3/0 | 14/42/0 |
| MPA | 1/11/0 | 5/5/2 | 10/10/4 | 8/0/0 | 24/26/6 |
| DE | 9/2/1 | 7/3/2 | 13/8/3 | 8/0/0 | 37/13/6 |
| HBA | 10/2/0 | 10/2/0 | 19/5/0 | 8/0/0 | 47/9/0 |
| PSO | 9/3/0 | 9/3/0 | 18/5/1 | 8/0/0 | 44/11/1 |
| GWO | 12/0/0 | 12/0/0 | 20/4/0 | 8/0/0 | 52/4/0 |
| WOA | 12/0/0 | 12/0/0 | 24/0/0 | 8/0/0 | 56/0/0 |

better. Adding an eigen-frame exchange to an HBA–MPA hybrid does not close the gap to a cooperative model of state-of-the-art evolutionary algorithms.

### *7.5. Computational cost*

EIG-MPHBS and MPHBS-R used identical evaluation counts on every one of their paired runs, as Proposition 2 states. Table 7 reports the run times. The eigen frame adds 3.2–5.5% to the median run time of MPHBS-R, and EIG-MPHBS is faster than MPHBS, whose SARSA bookkeeping costs more than the eigendecomposition. The classical algorithms are faster per run because their iterations are simpler; with expensive objectives, where evaluations dominate, the differences in algorithmic overhead become negligible.

## 8. Discussion

*What the eigen frame achieves.* The exchange step of HBA–MPA hybrids is where the two populations combine their information, and it was written for separable problems. Moving it into the eigen frame of the pooled populations makes it invariant to rotations at no extra evaluation cost, and the results show that this matters: EIG-MPHBS is never significantly worse than MPHBS on any of 56 instances and is better on 14, with the largest gains on ill-conditioned rotated functions, on CEC2022 at $D = 20$ and on the FIR application. The change is local, a few lines in the mediator, and should transfer to other hybrids that assemble solutions coordinate by coordinate.

*What it does not achieve.* The frame helps unimodal or moderately multimodal problems with coupled variables, where the population covariance reflects the basin; it does not help highly multimodal rotated functions such as f15, and it costs a little on separable multimodal ones such as f3 and f4, where half of the exchanges no longer follow the separable axes. Phase 1 remains coordinate-wise, which is why the gap to CMA-ES and L-SHADE on rotated functions persists. The comparison with EA4eig confirms that this is an improvement of a hybrid within its class, not a new state of the art.

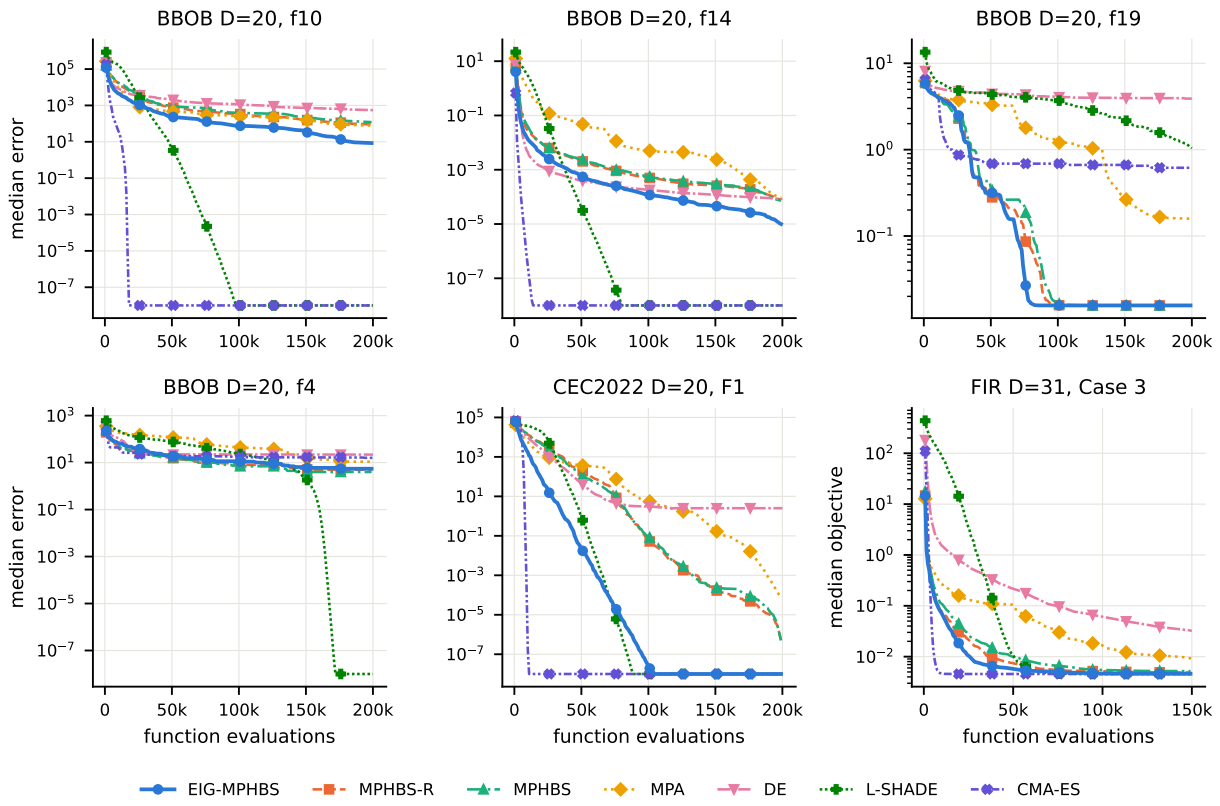



Figure 3: Median best-so-far error over 30 runs against function evaluations (FIR: objective value), log scale. The CEC2022 and FIR instances are those with the strongest per-instance effect of EIG-MPHBS against MPHBS-R; BBOB f4 is the held-out instance where it is significantly worse.

*Practical guidance.* For practitioners who use HBA–MPA hybrids, for example because they are already part of an application pipeline, EIG-MPHBS is a safe replacement for MPHBS: same budget, similar run time, never worse in our tests and often better. When the choice of algorithm is open, L-SHADE and CMA-ES remain the stronger default on problems of this size, and the classical metaheuristics PSO, GWO and WOA are clearly weaker than all three hybrids.

*Threats to validity.* (i) The confirmatory effect is small and sensitive to the handling of tied instances; the stronger effects come from instances that include the pilot instances. (ii) The value $q = 0.5$ was chosen from a pilot with two values; other values or an adaptive $q$ may perform differently. (iii) The six classical algorithms were added after the main analysis and are reported descriptively; they use standard settings, not tuned ones. (iv) The EA4eig comparison is unpaired and across implementations. (v) Run times were measured on a shared laptop. (vi) All problems have $D \leq 31$; for larger $D$ the $O(D^3)$ decomposition and the reliability of the covariance estimate from $2N$ points should be checked.

## 9. Conclusion

We identified the coordinate dependence of the dimension-wise exchange in the HBA–MPA hybrid MPHBS as a structural weakness on rotated problems and proposed EIG-

Table 6: EIG-MPHBS against the official CEC2022 runs of EA4eig at the competition budgets ($2 \times 10^5$ evaluations for $D = 10$, $10^6$ for $D = 20$; 30 runs each). Median final error (at or below $10^{-8}$ shown as 0) and Vargha–Delaney $A$ ($> 0.5$ favours EIG-MPHBS). Superscripts: two-sided Mann–Whitney test, Holm over the 12 functions of each dimension ($^+$ EIG-MPHBS better, $^-$ worse, $^\approx$ no significant difference). Unpaired and descriptive: hardware and implementation differ.

| | $D = 10$ | | | $D = 20$ | | |
|---|---|---|---|---|---|---|
| | EIG-MPHBS | EA4eig | $A$ | EIG-MPHBS | EA4eig | $A$ |
| F1 | $0^\approx$ | 0 | 0.50 | $0^\approx$ | 0 | 0.50 |
| F2 | $0^\approx$ | 0 | 0.46 | $47^-$ | 0 | 0.17 |
| F3 | 2.27e-7$^-$ | 0 | 0.13 | 7.77e-8$^-$ | 0 | 0.15 |
| F4 | $7.96^-$ | 0.995 | 0.01 | $34.8^-$ | 6.96 | 0.00 |
| F5 | $0^\approx$ | 0 | 0.50 | $14.1^-$ | 0 | 0.00 |
| F6 | $0.155^-$ | 5.41e-3 | 0.04 | $3.28^-$ | 0.106 | 0.01 |
| F7 | 2.36e-6$^-$ | 0 | 0.02 | $25.3^-$ | 2.3 | 0.03 |
| F8 | $0.16^-$ | 0.0467 | 0.27 | $20.6^-$ | 20.3 | 0.29 |
| F9 | $229^-$ | 186 | 0.00 | $181^-$ | 165 | 0.00 |
| F10 | $100^-$ | 100 | 0.11 | $100^\approx$ | 100 | 0.34 |
| F11 | $0^\approx$ | 0 | 0.50 | $300^-$ | 300 | 0.16 |
| F12 | $159^-$ | 146 | 0.00 | $240^-$ | 200 | 0.00 |

MPHBS, which performs the exchange in the eigenvector frame of the pooled populations in half of the iterations. The exchange becomes provably rotation invariant at no extra evaluation cost and a few percent of extra run time. In 18,840 runs on CEC2022, BBOB and FIR filter design, EIG-MPHBS improved on MPHBS on 14 instances and was never worse, passed a test fixed in advance on 28 held-out instances (narrowly), and outperformed its parent algorithms and four classical metaheuristics. It ranked third of eleven, behind L-SHADE and CMA-ES, and remained behind EA4eig at the CEC2022 budgets. Future work includes an adaptive choice of the frame probability, rotation-invariant versions of the Phase-1 moves, and tests at higher dimension.

## Data and code availability

Code, the design document (`experiments/ESWA_EIGEN_CAMPAIGN.md`), all 18,840 raw runs, the analyses and the scripts that generate every number, table and figure in this paper are available at `https://github.com/abinaya-g/la-kaie` [PENDING: campaign commit and release tag]. The EA4eig results are the official CEC2022 competition files, redistributed with their source in `data/external/cec2022_ea4eig/`.

## CRediT authorship contribution statement

**Abinaya G:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Table 7: Median wall-clock time per run in seconds (Python 3, NumPy, one core per run, 11 runs in parallel on a 12-core laptop; other jobs shared the machine, so times are indicative). Last row: median paired ratio of the run times of EIG-MPHBS and MPHBS-R, which use identical evaluation counts.

| Algorithm | C10 | C20 | B20 | FIR |
|---|---|---|---|---|
| EIG-MPHBS | 61.3 | 60.9 | 133.2 | 79.8 |
| MPHBS-R | 60.6 | 59.2 | 132.1 | 77.1 |
| MPHBS | 81.1 | 75.9 | 158.3 | 86.9 |
| HBA | 7.1 | 8.4 | 8.4 | 20.5 |
| MPA | 4.5 | 5.7 | 5.7 | 19.2 |
| L-SHADE | 19.4 | 11.6 | 23.4 | 24.9 |
| CMA-ES | 120.8 | 89.1 | 183.2 | 75.6 |
| DE | 17.6 | 18.2 | 18.3 | 28.4 |
| PSO | 4.9 | 6.1 | 5.9 | 19.3 |
| GWO | 5.7 | 7.2 | 7.0 | 18.9 |
| WOA | 5.5 | 6.6 | 6.4 | 17.7 |
| EIG-MPHBS / MPHBS-R | 1.055 | 1.033 | 1.032 | 1.032 |

## Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Declaration of generative AI and AI-assisted technologies in the writing process

[PENDING: author's decision on the declaration; Elsevier requires one if AI tools were used.]


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## Appendix A. Median errors per instance

Table A.8: Median final error over 30 runs on CEC2022 $D$=10 (FIR: objective value). Errors at or below $10^{-8}$ are shown as 0; lowest median in bold. Superscripts compare with EIG-MPHBS (paired Wilcoxon, Holm within benchmark and comparator): $^{+}$ EIG-MPHBS significantly better, $^{-}$ significantly worse, $^{\approx}$ no significant difference.

| | EIG-MPHBS | MPHBS-R | MPHBS | HBA | MPA | L-SHADE | CMA-ES | DE | PSO | GWO | WOA |
|---|---|---|---|---|---|---|---|---|---|---|---|
| F1 | **0** | **0**$^{\approx}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | 0.0805$^{+}$ | **0**$^{\approx}$ | 137$^{+}$ | 8.29e+03$^{+}$ |
| F2 | **0** | **0**$^{\approx}$ | 3.99$^{\approx}$ | 4.09$^{+}$ | **0**$^{\approx}$ | 3.99$^{\approx}$ | 3.99$^{\approx}$ | 8.92$^{+}$ | 8.92$^{+}$ | 10.4$^{+}$ | 76$^{+}$ |
| F3 | 2.27e-7 | 2.46e-8$^{\approx}$ | 9.39e-8$^{\approx}$ | 5.75e-5$^{+}$ | 2.98e-5$^{+}$ | **0**$^{-}$ | **0**$^{-}$ | **0**$^{-}$ | **0**$^{\approx}$ | 0.131$^{+}$ | 34.8$^{+}$ |
| F4 | 7.96 | 8.95$^{\approx}$ | 8.95$^{\approx}$ | 15.9$^{+}$ | 6.96$^{\approx}$ | 2.98$^{-}$ | **0.995**$^{-}$ | 9.79$^{+}$ | 10.4$^{\approx}$ | 14.1$^{+}$ | 39$^{+}$ |
| F5 | **0** | **0**$^{\approx}$ | **0**$^{\approx}$ | 0.544$^{+}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | **0**$^{+}$ | **0**$^{+}$ | 1.36$^{+}$ | 402$^{+}$ |
| F6 | 0.155 | 0.178$^{\approx}$ | 0.169$^{\approx}$ | 1.42e+03$^{+}$ | **0.138**$^{\approx}$ | 0.311$^{+}$ | 0.49$^{+}$ | 2.56$^{+}$ | 3.35e+03$^{+}$ | 2.9e+03$^{+}$ | 2.25e+03$^{+}$ |
| F7 | 2.36e-6 | 4.83e-6$^{\approx}$ | 0.995$^{\approx}$ | 21$^{+}$ | 0.995$^{\approx}$ | **1.63e-6**$^{\approx}$ | 1.62$^{\approx}$ | 0.667$^{\approx}$ | 20.6$^{+}$ | 23.3$^{+}$ | 96.2$^{+}$ |
| F8 | 0.16 | 0.18$^{\approx}$ | 0.232$^{\approx}$ | 21$^{+}$ | **0.138**$^{\approx}$ | 0.669$^{\approx}$ | 17.3$^{+}$ | 0.763$^{+}$ | 20.7$^{+}$ | 24.5$^{+}$ | 32.4$^{+}$ |
| F9 | **229** | **229**$^{\approx}$ | **229**$^{\approx}$ | 230$^{+}$ | **229**$^{\approx}$ | **229**$^{\approx}$ | **229**$^{\approx}$ | **229**$^{\approx}$ | 230$^{+}$ | 231$^{+}$ | 388$^{+}$ |
| F10 | 100 | 100$^{\approx}$ | 100$^{\approx}$ | 210$^{+}$ | 100$^{\approx}$ | **100**$^{\approx}$ | 100$^{\approx}$ | 100$^{+}$ | 210$^{+}$ | 207$^{+}$ | 253$^{+}$ |
| F11 | **0** | **0**$^{\approx}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | **0**$^{+}$ | 178$^{+}$ | 130$^{+}$ | 279$^{+}$ |
| F12 | 159 | **159**$^{\approx}$ | 160$^{\approx}$ | 183$^{+}$ | 160$^{\approx}$ | 161$^{\approx}$ | 163$^{+}$ | 164$^{+}$ | 171$^{+}$ | 164$^{+}$ | 233$^{+}$ |

Table A.9: Median final error over 30 runs on CEC2022 $D$=20 (FIR: objective value). Errors at or below $10^{-8}$ are shown as 0; lowest median in bold. Superscripts compare with EIG-MPHBS (paired Wilcoxon, Holm within benchmark and comparator): $^{+}$ EIG-MPHBS significantly better, $^{-}$ significantly worse, $^{\approx}$ no significant difference.

| | EIG-MPHBS | MPHBS-R | MPHBS | HBA | MPA | L-SHADE | CMA-ES | DE | PSO | GWO | WOA |
|---|---|---|---|---|---|---|---|---|---|---|---|
| F1 | **0** | 2.18e-7$^{+}$ | 2.82e-7$^{+}$ | 2.69e-8$^{+}$ | 5.13e-5$^{+}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | 2.51$^{+}$ | 2.77e-8$^{+}$ | 3.97e+03$^{+}$ | 3.30e4$^{+}$ |
| F2 | 44.9 | 49.1$^{+}$ | 44.9$^{\approx}$ | 49.1$^{\approx}$ | 44.9$^{\approx}$ | 49.1$^{\approx}$ | **44.9**$^{\approx}$ | 49.1$^{+}$ | 54.4$^{+}$ | 93$^{+}$ | 653$^{+}$ |
| F3 | 1.72e-5 | 5.03e-6$^{\approx}$ | 6.93e-6$^{\approx}$ | 1.09$^{+}$ | 4.68e-3$^{+}$ | **0**$^{-}$ | **0**$^{-}$ | 2.33e-5$^{\approx}$ | 0.556$^{+}$ | 2.29$^{+}$ | 71.7$^{+}$ |
| F4 | 35.3 | 36.8$^{\approx}$ | 30.3$^{\approx}$ | 50.7$^{+}$ | 38.8$^{\approx}$ | 5.97$^{-}$ | **3.98**$^{-}$ | 19.5$^{-}$ | 38$^{\approx}$ | 44.3$^{+}$ | 148$^{+}$ |
| F5 | 16.3 | 15.4$^{\approx}$ | 26.5$^{\approx}$ | 161$^{+}$ | 14.9$^{\approx}$ | **0**$^{-}$ | **0**$^{-}$ | 0.499$^{-}$ | 8.15$^{\approx}$ | 158$^{+}$ | 2.32e+03$^{+}$ |
| F6 | 12.2 | 16.8$^{\approx}$ | 24$^{\approx}$ | 2.46e+03$^{+}$ | 10.6$^{\approx}$ | **0.489**$^{-}$ | 52.5$^{+}$ | 3.21e+03$^{+}$ | 2.45e+03$^{+}$ | 2.08e4$^{+}$ | 7.11e7$^{+}$ |
| F7 | 31.1 | 29.8$^{\approx}$ | 27.4$^{\approx}$ | 69.1$^{+}$ | 24.9$^{-}$ | **20.3**$^{-}$ | 21.6$^{-}$ | 24.2$^{\approx}$ | 29.2$^{\approx}$ | 51.8$^{+}$ | 200$^{+}$ |
| F8 | 21 | 21$^{\approx}$ | 21.2$^{\approx}$ | 24.9$^{+}$ | 21.1$^{\approx}$ | 20.9$^{\approx}$ | **20.8**$^{\approx}$ | 22.2$^{+}$ | 41.3$^{+}$ | 28.3$^{+}$ | 58.2$^{+}$ |
| F9 | 181 | 181$^{+}$ | 181$^{+}$ | 181$^{+}$ | 181$^{+}$ | **181**$^{-}$ | **181**$^{-}$ | 181$^{+}$ | 218$^{+}$ | 208$^{+}$ | 443$^{+}$ |
| F10 | 100 | 100$^{\approx}$ | 100$^{\approx}$ | 839$^{+}$ | 101$^{+}$ | **100**$^{\approx}$ | 100$^{\approx}$ | 101$^{+}$ | 292$^{+}$ | 912$^{+}$ | 3.26e+03$^{+}$ |
| F11 | 300 | 300$^{\approx}$ | 300$^{\approx}$ | 300$^{\approx}$ | 400$^{+}$ | **300**$^{-}$ | 300$^{\approx}$ | 303$^{\approx}$ | 966$^{+}$ | 748$^{+}$ | 2.9e+03$^{+}$ |
| F12 | 243 | 239$^{\approx}$ | 241$^{\approx}$ | 357$^{+}$ | 241$^{-}$ | **234**$^{-}$ | 236$^{-}$ | 246$^{+}$ | 285$^{+}$ | 261$^{+}$ | 504$^{+}$ |

Table A.10: Median final error over 30 runs on BBOB $D$=20 (FIR: objective value). Errors at or below $10^{-8}$ are shown as 0; lowest median in bold. Superscripts compare with EIG-MPHBS (paired Wilcoxon, Holm within benchmark and comparator): $^{+}$ EIG-MPHBS significantly better, $^{-}$ significantly worse, $^{\approx}$ no significant difference.

| | EIG-MPHBS | MPHBS-R | MPHBS | HBA | MPA | L-SHADE | CMA-ES | DE | PSO | GWO | WOA |
|---|---|---|---|---|---|---|---|---|---|---|---|
| f1 | **0** | **0**$^{\approx}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | 1.19$^{+}$ | 0.536$^{+}$ | 15.3$^{+}$ |
| f2 | **0** | **0**$^{\approx}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | 3.77e-4$^{+}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | 0.295$^{+}$ | 3.46e+03$^{+}$ | 1.50e4$^{+}$ | 1.78e5$^{+}$ |
| f3 | 1.99 | 0.995$^{\approx}$ | 0.995$^{\approx}$ | 35.3$^{+}$ | 5.47$^{+}$ | **0**$^{-}$ | 8.95$^{+}$ | 13.1$^{+}$ | 19$^{+}$ | 45.6$^{+}$ | 337$^{+}$ |
| f4 | 5.47 | 4.97$^{-}$ | 3.98$^{\approx}$ | 45.3$^{+}$ | 10.9$^{+}$ | **0**$^{-}$ | 15.9$^{+}$ | 21.9$^{+}$ | 17.9$^{+}$ | 66.2$^{+}$ | 357$^{+}$ |
| f5 | **0** | **0**$^{\approx}$ | **0**$^{\approx}$ | 92.1$^{+}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | 30.8$^{+}$ | 10.8$^{+}$ | 106$^{+}$ |
| f6 | 3.26e-8 | 1.78e-6$^{+}$ | 3.37e-6$^{+}$ | 5.41e-4$^{+}$ | 1.06e-4$^{+}$ | **0**$^{-}$ | **0**$^{-}$ | **0**$^{\approx}$ | 114$^{+}$ | 40.8$^{+}$ | 1.19e+03$^{+}$ |
| f7 | 4.31 | 3.59$^{\approx}$ | 2.62$^{\approx}$ | 7.55$^{+}$ | 2.05$^{\approx}$ | **0**$^{-}$ | **0**$^{-}$ | 2.62$^{\approx}$ | 13.1$^{+}$ | 14.5$^{+}$ | 183$^{+}$ |
| f8 | 3.64e-7 | 4.22e-4$^{+}$ | 1.84e-3$^{+}$ | 3.31$^{+}$ | 10.7$^{+}$ | **0**$^{-}$ | **0**$^{-}$ | 17.2$^{+}$ | 12.8$^{+}$ | 90.7$^{+}$ | 2.82e+03$^{+}$ |
| f9 | 7.03e-3 | 0.798$^{+}$ | 1.48$^{+}$ | 6.41$^{+}$ | 4.57$^{+}$ | **0**$^{-}$ | **0**$^{-}$ | 17.7$^{+}$ | 16.9$^{+}$ | 22$^{+}$ | 107$^{+}$ |
| f10 | 8.47 | 89.9$^{+}$ | 121$^{+}$ | 829$^{+}$ | 77.3$^{+}$ | **0**$^{-}$ | **0**$^{-}$ | 550$^{+}$ | 7.09e+03$^{+}$ | 1.15e4$^{+}$ | 2.28e5$^{+}$ |
| f11 | **0** | 3.36e-3$^{+}$ | 8.62e-3$^{+}$ | 1.09$^{+}$ | 4.60e-4$^{+}$ | **0**$^{\approx}$ | **0**$^{\approx}$ | 2.65$^{+}$ | 1.65$^{+}$ | 40.4$^{+}$ | 111$^{+}$ |
| f12 | 4.65 | 3.47$^{\approx}$ | 4.35$^{\approx}$ | 3.45$^{\approx}$ | 5.16$^{\approx}$ | **0**$^{-}$ | **0**$^{-}$ | 1.96$^{\approx}$ | 7.93e4$^{+}$ | 2.46e5$^{+}$ | 2.85e7$^{+}$ |
| f13 | 0.774 | 2.82$^{\approx}$ | 2.86$^{\approx}$ | 3$^{+}$ | 1.13$^{\approx}$ | **0**$^{-}$ | **0**$^{-}$ | 2.04$^{+}$ | 200$^{+}$ | 179$^{+}$ | 858$^{+}$ |
| f14 | 9.73e-6 | 7.96e-5$^{+}$ | 7.21e-5$^{+}$ | 7.79e-5$^{+}$ | 9.53e-5$^{+}$ | **0**$^{-}$ | **0**$^{-}$ | 8.40e-5$^{+}$ | 1.58$^{+}$ | 0.26$^{+}$ | 8.69$^{+}$ |
| f15 | 54.2 | 58.7$^{\approx}$ | 51.2$^{\approx}$ | 70.1$^{+}$ | 37.3$^{-}$ | 4.13$^{-}$ | **2.98**$^{-}$ | 19$^{-}$ | 57.1$^{\approx}$ | 63.9$^{+}$ | 359$^{+}$ |
| f16 | 1.84 | 1.99$^{\approx}$ | 1.92$^{\approx}$ | 6.02$^{+}$ | 1.39$^{\approx}$ | 3.83$^{+}$ | **0.0403**$^{-}$ | 9.82$^{+}$ | 3.21$^{+}$ | 1.28$^{\approx}$ | 15.8$^{+}$ |
| f17 | 0.488 | 0.592$^{\approx}$ | 0.448$^{\approx}$ | 1.38$^{+}$ | 0.271$^{\approx}$ | **0**$^{-}$ | **0**$^{-}$ | 0.0333$^{-}$ | 0.753$^{\approx}$ | 0.918$^{\approx}$ | 7.25$^{+}$ |
| f18 | 1.04 | 1.15$^{\approx}$ | 0.86$^{\approx}$ | 4.83$^{+}$ | 0.997$^{\approx}$ | 4.66e-8$^{-}$ | **0**$^{-}$ | 0.241$^{-}$ | 1.62$^{\approx}$ | 2.31$^{+}$ | 26.9$^{+}$ |
| f19 | **0.0157** | 0.0157$^{+}$ | 0.0157$^{+}$ | 0.023$^{+}$ | 0.159$^{+}$ | 1.04$^{+}$ | 0.616$^{+}$ | 3.9$^{+}$ | 3.65$^{+}$ | 3.32$^{+}$ | 0.247$^{+}$ |
| f20 | 1 | 0.987$^{\approx}$ | 0.972$^{\approx}$ | 1.37$^{+}$ | 1.08$^{\approx}$ | **0.221**$^{-}$ | 1.15$^{\approx}$ | 0.898$^{\approx}$ | 0.75$^{-}$ | 1.52$^{+}$ | 5.68$^{+}$ |
| f21 | 2.04 | 1.92$^{\approx}$ | 1.8$^{\approx}$ | 2.82$^{\approx}$ | **0.346**$^{-}$ | 1.8$^{-}$ | 0.93$^{-}$ | 2.04$^{\approx}$ | 2.52$^{\approx}$ | 5.27$^{\approx}$ | 27.5$^{+}$ |
| f22 | 8.74 | 8.74$^{\approx}$ | 8.74$^{\approx}$ | 8.74$^{\approx}$ | **1.32**$^{-}$ | 7.27$^{\approx}$ | 1.91$^{-}$ | 2.59$^{\approx}$ | 8.65$^{\approx}$ | 7.4$^{\approx}$ | 36.2$^{+}$ |
| f23 | **0.0735** | 0.0948$^{\approx}$ | 0.0802$^{\approx}$ | 1.72$^{+}$ | 0.106$^{\approx}$ | 0.569$^{+}$ | 1.61$^{+}$ | 1.9$^{+}$ | 1.51$^{+}$ | 1.82$^{+}$ | 1.46$^{+}$ |
| f24 | 62.5 | 58$^{\approx}$ | 62.1$^{\approx}$ | 89.9$^{+}$ | 36.8$^{-}$ | 32.6$^{-}$ | **23.5**$^{-}$ | 102$^{+}$ | 109$^{+}$ | 137$^{+}$ | 182$^{+}$ |

Table A.11: Median final error over 30 runs on FIR $D$=31 (FIR: objective value). Errors at or below $10^{-8}$ are shown as 0; lowest median in bold. Superscripts compare with EIG-MPHBS (paired Wilcoxon, Holm within benchmark and comparator): $^{+}$ EIG-MPHBS significantly better, $^{-}$ significantly worse, $^{\approx}$ no significant difference.

| | EIG-MPHBS | MPHBS-R | MPHBS | HBA | MPA | L-SHADE | CMA-ES | DE | PSO | GWO | WOA |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Case 1 | 4.64e-3 | 4.98e-3$^{+}$ | 5.01e-3$^{+}$ | 0.0113$^{+}$ | 9.09e-3$^{+}$ | 4.56e-3$^{-}$ | **4.56e-3**$^{-}$ | 0.0306$^{+}$ | 0.0982$^{+}$ | 0.165$^{+}$ | 0.62$^{+}$ |
| Case 2 | 0.0207 | 0.0209$^{\approx}$ | 0.0209$^{\approx}$ | 0.0351$^{+}$ | 0.0302$^{+}$ | 0.0188$^{-}$ | **0.017**$^{-}$ | 0.0662$^{+}$ | 0.144$^{+}$ | 0.181$^{+}$ | 0.705$^{+}$ |
| Case 3 | 4.61e-3 | 4.90e-3$^{+}$ | 5.19e-3$^{+}$ | 0.0118$^{+}$ | 9.29e-3$^{+}$ | 4.56e-3$^{-}$ | **4.56e-3**$^{-}$ | 0.0321$^{+}$ | 0.0685$^{+}$ | 0.185$^{+}$ | 0.925$^{+}$ |
| Case 4 | 0.0199 | 0.0213$^{\approx}$ | 0.0213$^{+}$ | 0.0344$^{+}$ | 0.0279$^{+}$ | 0.017$^{-}$ | **0.017**$^{-}$ | 0.0616$^{+}$ | 0.141$^{+}$ | 0.166$^{+}$ | 0.96$^{+}$ |
| Case 5 | 1.40e-3 | 1.42e-3$^{\approx}$ | 1.73e-3$^{+}$ | 5.62e-3$^{+}$ | 4.31e-3$^{+}$ | 9.68e-4$^{-}$ | **8.88e-4**$^{-}$ | 0.0671$^{+}$ | 1.87$^{+}$ | 0.0837$^{+}$ | 0.943$^{+}$ |
| Case 6 | 6.17e-3 | 6.21e-3$^{\approx}$ | 5.87e-3$^{\approx}$ | 9.94e-3$^{+}$ | 9.74e-3$^{+}$ | 4.70e-3$^{-}$ | **4.70e-3**$^{-}$ | 0.069$^{+}$ | 0.226$^{+}$ | 0.0889$^{+}$ | 0.948$^{+}$ |
| Case 7 | 1.37e-3 | 1.36e-3$^{\approx}$ | 1.30e-3$^{\approx}$ | 5.21e-3$^{+}$ | 7.21e-3$^{+}$ | 8.15e-4$^{-}$ | **6.12e-4**$^{-}$ | 0.0358$^{+}$ | 2.69$^{+}$ | 0.0818$^{+}$ | 0.533$^{+}$ |
| Case 8 | 3.23e-3 | 3.81e-3$^{\approx}$ | 3.89e-3$^{+}$ | 0.0119$^{+}$ | 8.38e-3$^{+}$ | 1.81e-3$^{-}$ | **1.35e-3**$^{-}$ | 0.0117$^{+}$ | 2.58$^{+}$ | 0.096$^{+}$ | 0.48$^{+}$ |

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